MSPRF Sponsoring Scientist Statement

Proposed Sponsoring Scientific Advisor:
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This note is a sponsoring scientist statement for Edward Varvak to receive an NSF postdoctoral fellowship at MIT, starting 2026-2027.

Let me begin by discussing the compatibility of myself as sponsoring scientist with Varvak's research focus and proposal. Varvak's interests involve hyperkahler geometry and Lagrangian fibrations. He has written a series of papers studying Hodge-theoretic constraints on the fibers of a compact Lagrangian fibration; specifically, there are strong conditions of the Mumford-Tate group of the generic fiber of such families. He has a number of plans to study these constraints further and examine the noncompact setting. These questions overlap with my own recent research work, in which me and my collaborators are interested in studying special properties of the topology of Lagrangian fibrations (when compared to general abelian fibrations). As a result, I expect there will be many points of common research contact.

Furthermore, MIT (and the Boston area overall) is a natural fit for these interests. There are several geometers with Hodge-theoretic interest in the greater Boston area (notably Mihnea Popa)

In terms of the logistics of sponsorship, I expect to be regularly available during the fellowship period. In particular, I have no long-term travel plans during this period. If he receives the fellowship, I would meet him on a regular basis to discuss his research progress and any related questions that arise. If he's interested, he could help organize algebraic geometry seminar and the preprint seminar. Beyond the research aspects of his training, MIT offers ample opportunities for non-research professional development; most notably, the department offers an intensive teacher training program for new postdocs, which he would be encouraged to participate in, as well as workshops in grant-writing and undergraduate advising.